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EXPERIMENT – 24

MATLAB Simulation of Passband Transmission with BER Analysis Under Multipath Fading Channels

PROGRAM:

```
clc;
clear;
close all;
% Parameters
N = 1000;
bits = randi([0 1],N,1);
% BPSK Modulation
tx = 2*bits - 1;
% Multipath Fading Channel
h = [1 0.5 0.3];
fade = filter(h,1,tx);
% Add AWGN
rx = awgn(fade,10,'measured');
% Detection
bits_rx = rx > 0;
% BER Calculation
[~, BER] = biterr(bits,bits_rx);
fprintf('BER = %f\n',BER);
% Plots
figure;
subplot(3,1,1)
plot(tx)
title('Transmitted BPSK Signal')
grid on
```

```
subplot(3,1,2)
plot(rx)
title('Received Signal (Multipath + AWGN)')
grid on
subplot(3,1,3)
stem(bits(1:50),'b')
hold on
stem(bits_rx(1:50),'r')
title(['BER = ' num2str(BER)])
legend('Original','Detected')
grid on
```

OUTPUT:

